

GCP Security Monitoring and Logging Lab

Cloud Security Engineering Project Report

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Role Target: Cloud Security Engineer

Platform: Google Cloud Platform (GCP)

Project Type: Cloud Security Monitoring, Audit Logging, Detection Engineering, and IAM Security Review

Project Status: Completed

Environment: Personal GCP Project Environment

Executive Summary

This project demonstrates the implementation of cloud-native security monitoring and logging controls within Google Cloud Platform (GCP). The objective was to establish visibility into cloud activities, generate actionable security telemetry, create custom detection mechanisms, configure monitoring alerts, and assess Identity and Access Management (IAM) security posture.

The lab simulated key responsibilities of a Cloud Security Engineer, including audit log analysis, detection engineering, monitoring configuration, cloud resource monitoring, and access control assessment.

During the project, Google Cloud Audit Logs were analyzed, a Compute Engine virtual machine was deployed, custom Log-Based Metrics were created, Cloud Monitoring alert policies were configured, and IAM permissions were reviewed against cloud security best practices.

The project successfully demonstrated how cloud security teams can collect, monitor, detect, and respond to infrastructure activities using native GCP security services.

Project Objectives

The primary objectives of this project were:

- Configure cloud security monitoring capabilities within GCP.
- Enable and validate cloud audit logging.
- Investigate cloud resource activity through Cloud Logging.
- Create custom Log-Based Metrics for security monitoring.
- Configure Cloud Monitoring alert policies.
- Review IAM permissions and identify security risks.
- Apply cloud security best practices and least privilege principles.
- Document findings and recommendations using professional security reporting methodologies.

Project Architecture

Cloud Environment Components:

- Google Cloud Project
- Cloud Logging
- Cloud Audit Logs
- Compute Engine
- Log-Based Metrics
- Cloud Monitoring
- IAM & Admin

Security Monitoring Workflow

Cloud Activity



Cloud Audit Logs



Cloud Logging



Custom Log-Based Metrics



Cloud Monitoring



Alert Policies



Security Investigation

Lab Environment

Component	Purpose
Google Cloud Project	Dedicated monitoring environment
Cloud Logging	Centralized log collection
Cloud Audit Logs	Administrative activity tracking
Compute Engine VM	Activity generation and monitoring
Log-Based Metrics	Detection engineering
Cloud Monitoring	Alert generation
IAM & Admin	Access review and security assessment

Phase 1: Project Creation

A dedicated Google Cloud project named:

gcp-security-monitoring-lab

was created to isolate monitoring activities and security testing.

Outcome

- Isolated cloud environment established.
- Billing enabled.
- Security services prepared for deployment.

Phase 2: Security Service Enablement

The following APIs were enabled:

- Cloud Logging API
- Cloud Monitoring API
- Compute Engine API
- Cloud Asset API
- Security Command Center API

Security Importance

These APIs provide visibility, monitoring, auditing, inventory management, and cloud security functionality.

Outcome

Required cloud security services successfully activated.

Phase 3: Security Command Center Assessment

An attempt was made to access Security Command Center (SCC).

Observation

Security Command Center required an organizational account and was unavailable within the personal GCP environment.

Security Assessment

This limitation is expected in personal projects and does not prevent implementation of effective cloud monitoring.

Alternative Controls Used

- Cloud Audit Logs
- Cloud Logging
- Log-Based Metrics
- Cloud Monitoring
- IAM Reviews

Outcome

Cloud-native monitoring controls were used as an alternative security monitoring strategy.

Phase 4: Cloud Audit Log Analysis

Cloud Logging was used to investigate audit activity.

Query Used

resource.type="audited_resource"

Events Observed

- EnableService
- GetOperation
- Service Usage Events
- Cloud OS Config Operations
- Billing Configuration Activity

Security Value

Audit logs provide:

- Administrative accountability
- Change tracking
- Security investigations
- Incident response evidence
- Compliance reporting

Outcome

Successfully validated audit log collection and visibility.

Phase 5: Compute Engine Monitoring

A test virtual machine was deployed.

VM Configuration

Setting	Value
Name	monitoring-test-vm
Machine Type	e2-micro
Region	asia-south1
Operating System	Debian Linux

Purpose

Generate monitored cloud activity for logging and alerting validation.

Outcome

Successfully generated cloud infrastructure activity for monitoring analysis.

Phase 6: Detection Engineering

A custom Log-Based Metric was created.

Metric Name

vm_activity_metric

Metric Type

Counter Metric

Detection Logic

Monitor Compute Engine-related audit activity.

Security Benefits

- Converts logs into measurable security events.
- Enables automated monitoring.
- Supports alert creation.
- Improves visibility into infrastructure changes.

Outcome

Custom detection metric successfully implemented.

Phase 7: Security Alerting

A Cloud Monitoring Alert Policy was configured.

Policy Name

GCE VM Activity Alert

Data Source

vm_activity_metric

Configuration

Setting	Value
Condition Type	Threshold

Setting	Value
Trigger	Any Time Series Cross Threshold
Threshold	Greater Than 0

Alert Workflow

Cloud Activity

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Audit Logs

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Log-Based Metric

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Cloud Monitoring

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Security Alert

Outcome

Successfully implemented automated cloud activity detection and alerting.

Phase 8: IAM Security Assessment

IAM permissions were reviewed to identify security risks.

Finding 1

User Account

Role:

Owner

Risk

Owner permissions provide unrestricted access to all project resources.

Recommendation

Apply Principle of Least Privilege.

Assign role-specific permissions whenever possible.

Finding 2

Compute Engine Default Service Account

Role:

Editor

Risk

Editor permissions grant broad access across multiple services.

Compromise of the service account may result in excessive privilege exposure.

Recommendation

Replace Editor with dedicated least-privilege service accounts.

Use workload-specific IAM roles.

Security Findings Summary

ID	Finding	Risk Level
F-01	Owner Role Assignment	Medium
F-02	Default Service Account Uses Editor Role	High
F-03	Security Command Center Unavailable in Personal Environment	Informational

Security Recommendations

Logging

- Enable long-term log retention.
- Export logs to centralized SIEM solutions.

Monitoring

- Expand monitoring coverage to storage, IAM, networking, and API services.
- Configure high-risk activity alerts.

IAM

- Remove unnecessary administrative permissions.
- Replace default service account roles.
- Implement least privilege principles.

Governance

- Implement security baselines.
 - Conduct regular IAM reviews.
 - Enable organization-level security services when available.
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Skills Demonstrated

This project demonstrates practical experience with:

- Google Cloud Platform (GCP)
 - Cloud Security Monitoring
 - Audit Logging
 - Cloud Logging
 - Cloud Monitoring
 - Detection Engineering
 - Security Alerting
 - Compute Engine Security
 - IAM Security Reviews
 - Security Operations
 - Incident Detection
 - Cloud Governance
 - Security Documentation
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Business Impact

This implementation improves:

- Security visibility
 - Threat detection capabilities
 - Cloud activity monitoring
 - Incident investigation readiness
 - IAM security posture
 - Operational security maturity
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Conclusion

This project successfully implemented a cloud-native security monitoring solution using Google Cloud Platform services. Through audit log analysis, detection engineering, monitoring configuration, and IAM security assessment, the environment achieved improved visibility into cloud operations and security events.

The project reflects practical Cloud Security Engineer responsibilities and demonstrates hands-on experience with security monitoring, logging, detection, and access control management within a modern cloud environment.

The successful completion of this project validates foundational cloud security engineering skills and provides a strong portfolio artifact for entry-level Cloud Security Engineer opportunities.